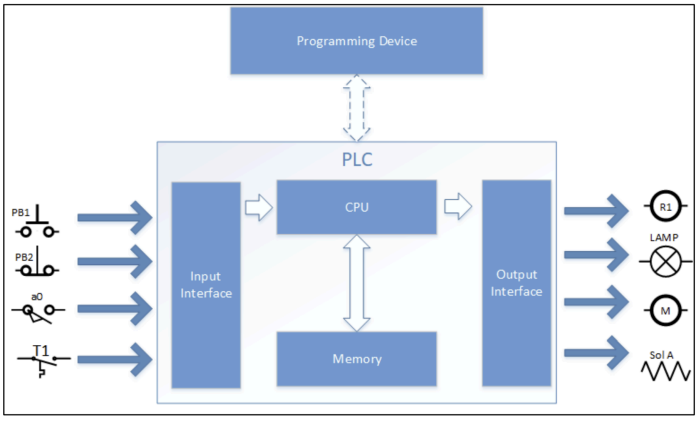
Programmable Logic Controllers

Programmable Logic Controllers, PLCs, are signal processing units that replace large and complicated relay control systems.

The principle of PLCs is that they have a CPU that recieves input signals, and using the memory and program on the CPU, outputs specific signals to the output interfaces.



This reduces the need for hardwiring; only required for connecting the PLC to the input and output devices. Relays, timers and counters are also built into the PLC instead and can have an unlimited number of them. They are also much more flexible as their program can be changed without changing the wiring. They are also much more compact.

However, they are quite expensive compared to simple relay systems, and the lack of standardization between PLCs makes it difficult to standardize programs.

PLC Components

- The CPU reads input signals and does signal processing on the signals, like boolean logic, arithmetic, using the memory to store program data, variables, and operational values used.
- To program the PLC, we use a programming device to upload and debug the program, but once the program is uploaded, it does not require the programming device anymore to run the same program multiple times.
- Inputs are connected to the interface where signals are transferred to the CPU via opto-isolation; i.e. using light, to prevent overloading the PLC.
- likewise, outputs are connected to the output interface, using a relay to transmit signals but not power.

Program Execution

- during execution, the PLC performs a scan cycle; the time taken to first read all the inputs, then the time taken to process the inputs and operate on them, and then the time taken to output the respective signals.

Omron PLC

- For this module, we use the Omron PLC with Sysmac studio as the programming device, using ladder diagrams as our programming language.

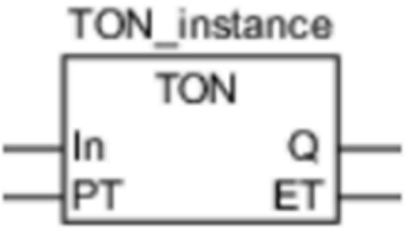
PLC Variables

- there are 3 types of variables used, Internal/Local, External/Global, and System variables
- global variables mean that the variable can be access by other programs, usually having a ‘g’ prefix in front. system variables are used by the plc internally to record states of other properties, but are not discussed in the module.

Ladder Programming Components

- Inputs (NO/NC)
  - inputs are represented as relay switches, with two bars that can be either normally open or closed.
- Outputs
  - outputs are represented as relay coils, as circles. they can either refer to global variable outputs of the PLC, or new variables or block variables.

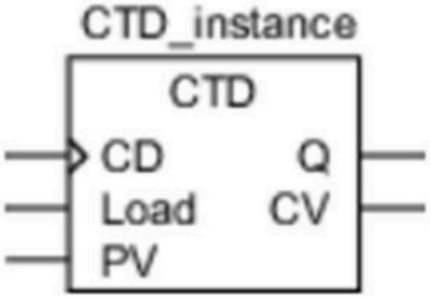
TON timer blocks



TON stands for timer ON delay, where in is the input signal which starts the timer when In is true. PT is the preset time which determines how long the timer will go for, ET is the elapsed time, and Q is the output that is true when PT = ET, or when the timer has fully elapsed.

Usually, the output Q is used as an input for other ladder rungs to activate other rungs.

CTD counters



CTD stands for counter down, meaning it counts down from a present number to 0.

CD is the counter input and recognises rising edges as a pulse, decrementing the counter by 1. Load signal is an input that resets the counter if true. PV is the preset value of the value it starts counting down from.

CV is the current count as an output and Q is the boolean output when the CTD has finished counting down.

Electro-pneumatic Control

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